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Domain: Tools and Techniques for Determining Malicious Activity

Episode: Email Analysis Tools

Email Analysis Tools

Objectives

- Explain a bit about email analysis
 - Impersonation
 - Headers
 - Embedded Links
 - Sender Policy Framework(SPF)
 - Authentication system used to verify allowed "Sending email" Servers
 - Is this server allowed to send email on behalf of xyz.com?
 - Uses SPF records, which is published with its DNS record
 - Manually inspect SPF record
 - ```
nslookup
set type=txt
acilearning.com
```
- 
- DomainKeys Identified Mail(DKIM)
  - Use of Digital Signatures to verify email integrity
    - Public/Private keys
      - Email recipient uses senders Public Key to validate digital signatures
- Domain-based Message Authentication, Reporting, and Conformance (DMARC)
  - Advertises that the sender domain is using SPF and/or DKIM
  - Tells recipient what to do with email that fails those auth methods
    - <https://dmarcian.com/dmarc-inspector/>
    - <https://datatracker.ietf.org/doc/html/rfc7489>